

Shear-wave crosswell tomography using distributed acoustic sensing: A breakthrough in geotechnical site characterization

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<https://doi.org/10.1190/tle-2025-1012>

Abstract

This study presents a novel approach to perform high-resolution S-wave crosswell tomography by combining distributed acoustic sensing (DAS) with a borehole-coupled SV-wave source. Linear DAS fibers, which exhibit optimal sensitivity to vertically polarized shear waves, were used to acquire dense seismic data sets with significantly improved spatial resolution compared to conventional geophones. The SV-wave source (BIS-SV) provides efficient signal generation and coupling, allowing accurate traveltimes picking and stable tomographic inversion results. Field experiments conducted at the ECC-SEL Svelvik CO₂ Field Lab demonstrated that DAS-based S-wave tomography can be performed with acquisition times of 3–4 min per shot depth and a twofold increase in spatial resolution compared to conventional methods. In contrast to P-wave tomography, S-wave results revealed distinct subsurface structures and allowed direct derivation of dynamic shear stiffness, a key parameter in geotechnical engineering. A low correlation ($R^2 = 0.196$) between P- and S-wave traveltimes confirmed that both methods provide complementary insights into subsurface properties. In addition, the DAS system eliminated the need for water-filled boreholes and mechanical geophone coupling, providing significant cost and logistical advantages. DAS seismic tomography adds significant value by providing continuous, high-resolution, cost-effective subsurface monitoring with minimal field effort, making it a transformative tool for modern geotechnical and environmental applications.

Introduction

Seismic tomography is a technique to image the earth's interior by using seismic waves that are either generated by earthquakes, vibrations, or explosive sources and provides valuable information about the heterogeneous nature of the subsurface across distances ranging from thousands of kilometers to a few meters (Rawlinson et al., 2014). As a common standard, P-wave tomography is widely applied as an integrated tool to predict high-resolution spatial continuity of lithological structures (Paasche et al., 2011), and it is used to delineate geological structures, map cavities and weak zones, monitor deep reservoirs and storage sites, and explore for oil and gas. P-wave tomography is also used in geotechnical engineering for soil characterization before infrastructure is built and to monitor time-dependent processes, such as cement soil

improvement. Additionally, it helps to image the foundations of buildings and bridges, as well as their underlying soil (Stewart, 1991; Rammah et al., 2022; Fichtner et al., 2024).

Crosswell seismic tomography is used to measure the 2D velocity field between two boreholes (Koedel et al., 2022). With its at least four times higher resolution, crosswell seismic tomography offers a solution to a lower resolution in conventional surface seismic data (Pereira and Jones, 2010). The technique is renowned for its accessibility, as the equipment is user-friendly, well-established, and readily available on the market. Furthermore, data processing is efficient and straightforward, making P-wave tomography the preferred choice for detailed geophysical crosswell studies.

In contrast to P-wave tomography, S-wave crosswell tomography is applied so far only in a relatively limited number of studies, predominantly motivated by academic interest. S-wave velocity measurements are only presented within the standards ASTM D4700 for downhole surveying and ASTM 4428 for cross-hole testing (ASTM, 2018, ASTM, 2025). This is mainly due to the typically limited number of sensors available — each of which must be properly coupled to the borehole wall — and the low energy of available sources, which often limits their ability to propagate over distances of tens of meters, the logistical challenges of field equipment deployment, and the time-intensive nature of data acquisition and processing (von Ketelhodt et al., 2017). For instance, the study present by von Ketelhodt et al. (2017) could only determine less than half of the S-wave traveltimes for inversion but needs more than two times the acquisition time for the S-wave survey and about two times more for processing of the S-wave data compared to the P-wave tomography. The advantages of S-wave over P-wave propagation are straightforward and include, in principle, a higher spatial resolution, a higher sensitivity to changes in material properties, the opportunity to map the unsaturated and saturated parts of the subsurface with the same resolution, and the direct link to soil dynamic properties, i.e., the shear stiffness (see Figure 1) at small strain rates. Shear stiffness plays a particularly important role in geotechnical engineering for understanding soil behavior under different loading conditions (Burland, 1989; Jamiolkowski et al., 1995; Atkinson, 2000). In Figure 1, the advantages and disadvantages of S-wave tomography compared with conventional P-wave tomography are shown.

Manuscript received 29 July 2025; revision received 24 October 2025; accepted 18 November 2025.

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Given these advantages, S-wave tomography should be more widely used in seismic investigations, particularly in applications requiring high-resolution imaging of lithological structures, fault zones, and geomechanical properties. Its ability to provide detailed insights into rock stiffness and structural integrity makes it an invaluable tool in earthquake hazard assessment, site characterization for construction projects, and subsurface monitoring for geothermal and carbon sequestration projects (e.g., Hussien and Karray, 2015). In addition to land-based vibrator systems, there are borehole sources on the market that can generate controlled P- and S-waves. Minto et al. (2012) and Nalonnil, Marion, and Minto (2013) demonstrate the use of a dual-wavefield downhole vibrator (Z-Trac) system that generates SH- and P-waves for crosswell applications. This dual wave-field borehole vibrator was used, e.g., at frequencies between 30 and 600 Hz to acquire a multicomponent data set for anisotropic full-waveform inversion (FWI) at the Wyoming CarbonSAFE project (Podgornova et al., 2023). This borehole source had two wave excitation components, 6 m apart. These were fired sequentially, whereas seismic waves were measured by three-component geophone receivers with a spatial sampling of 30 m. The strike direction of the SH-wave generation was unknown, and the data needed to be rotated to a crossline direction to obtain the best possible data quality. P-wave data were determined on the in-line component of the rotated geophone sensors. The application of anisotropic elastic FWI to crosswell seismic data demonstrated that FWI can use all complex crosswell waveforms.

Although P-wave tomography remains a widely used and accessible method due to its simplicity and fast data acquisition, integrating S-wave tomography into seismic studies can significantly enhance the accuracy and reliability of subsurface models. Future advancements in seismic acquisition techniques and data processing will further improve the effectiveness of S-wave tomography, solidifying its role as a modern tool in geophysics.

Significant effort has been dedicated to the development of advanced equipment that enables the acquisition of S-wave

borehole tomographic data in a more efficient and competitive manner. Traditionally, the acquisition of S-wave data has been significantly more time-consuming compared to standard P-wave tomography. This challenge primarily arises due to the complexity of generating reliable S-wave sources and the necessity of a multichannel receiver system capable of effectively capturing S-wave signals. Unlike P-waves, which are relatively easier to generate and detect, S-wave sources require specialized mechanisms to induce shear motion in the subsurface, ensuring accurate and consistent wave propagation. One of the key limitations of conventional multichannel receiver systems is the coupling to the borehole wall. Achieving stable and consistent coupling is crucial for accurate S-wave detection. Additionally, the limited number of receiver stations poses a challenge in achieving sufficient ray coverage over the entire borehole length. To overcome these limitations, advancements in borehole instrumentation and acquisition techniques have been developed. These innovations focus on improving the efficiency of S-wave generation, enhancing receiver coupling mechanisms, and increasing the number of active receiver stations to ensure comprehensive ray coverage. These improvements will ultimately make S-wave borehole tomography a more viable and competitive option for high-resolution subsurface imaging.

Additionally, S-wave data are key to calculating depth-dependent stiffness profiles as they reflect variations in shear modulus directly. These stiffness profiles are necessary for providing material parameters for various numerical investigations in geotechnical engineering. These parameters are strain-dependent, so realistic input values are required, such as the initial values of maximum stiffness and minimum damping at small strains. Although conventional laboratory tests are not suitable for such studies, reliable measurements of small-strain soil stiffness can only be made in the laboratory by means of triaxial tests with local strain measurements, bender elements, resonant column, and torsional shear tests. However, accurate soil parameters can only be obtained if the soil sample is undisturbed, and seismic field measurements can be used to obtain these initial values in a nondestructive way in naturally

occurring soils and can serve as an indirect test method for very small strain stiffness (Benz, 2007). For promising geotechnical investigations, a sensible combination of laboratory and field tests such as S-wave measurements is essential due to the different performance and information capabilities (Figure 2) of the abovementioned experimental methods at different strain ranges (e.g., Jamiolkowski et al., 1995; Tatsuoka et al., 1999; Meyer, Larkin, and Pender, 2005; Benz, 2007). The highest stiffness value G at the smallest strain obtained by seismic measurements is called G_0 .

S-wave Tomography



Figure 1. Advantages and disadvantages of S-wave tomography over conventional P-wave tomography.

Within the last decade, the development of distributed acoustic sensing (DAS) for seismic surveys has shown that there are several practical advantages over conventional sensors, such as higher spatial resolution of less than 1 m over distances beyond several tens of kilometers, repeatable measurements, and time savings (e.g., Boullenger et al., 2025). Moreover, DAS can produce favorable imaging outcomes in specific circumstances. For instance, the apparent amplitude of P-waves rises with the incidence angle. Similarly, the highest amplitudes of SV-waves are seen at the lowest incidence angles, as the particle movement is aligned with the fiber axis. DAS has been used for several years for vertical seismic profiling (VSP) acquisition in oil and gas exploration and for imaging plumes at Carbon Capture and Storage (CCS) sites (e.g., Daley et al., 2016; Halladay et al., 2018; Yurikov et al., 2022; Nakajima et al., 2022); only a few studies have been conducted on seismic crosswell tomography (e.g., Wuestefeld and Weinzierl, 2020). Also, many studies using DAS VSP P-wave data have been shown in the literature, but not much discussion has been given to S-wave DAS data (Willis et al., 2017; Willis, 2022). Wu et al. (2017) present a convincing comparison of DAS-VSP shot records and processes images of P- and S-wave sources with the corresponding geophone measurement. This paper shows the theoretical angular response of the fiber to P- and S-waves for a single point on the fiber and for the entire borehole. Koedel et al. (2024) analyze the signal-to-noise ratio of linear and helically wounded cables to conventional P-, SH-, and SV-wave borehole sources and find that good quality crosswell DAS data can be obtained for P- and SV-wave sources.

Using an SV-wave borehole source with DAS is a promising way to image and monitor the subsurface. It combines the benefits of shear waves and fiber optics to improve resolution and sensitivity to variations of subsurface properties. This combination potentially provides continuous, high-spatial-density measurements that outperform traditional geophones in terms of coverage and data resolution, providing a cost-effective solution for large-scale, real-time monitoring. The combination undoubtedly is a valuable tool for geotechnical soil characterization. The combined system can be additionally used in energy exploration, such as hydrocarbon reservoir monitoring, and

environmental studies, such as landslide detection and slope stability analysis.

In this paper, we present the first SV-wave crosswell survey using DAS technology and discuss the enormous time savings by using DAS for S-wave tomography, not only making S-wave tomography comparable to P-wave tomography in terms of speed and costs, but also providing direct access to high-resolution soil dynamic parameters required by the geotechnical community. The work was carried out within the framework of the EU-funded GeoInquire program to allow companies to access and to use existing test sites.

Field experiment and data acquisition

The crosswell S-wave tomography survey was conducted at the ECCSEL Svelvik CO₂ Field Lab (operated by SINTEF), located at the outlet of the Drammensfjord, Norway (Eliasson et al., 2018; SINTEF, 2025). The site is a small-scale field laboratory for rapid and cost-effective development and testing of technologies for quantitative geophysical monitoring for CO₂ injection experiments. The site includes a central injection well (S2), four monitoring wells (M1–M4), and the necessary infrastructure for CO₂ injection experiments (see Figure 3). All monitoring boreholes are PVC-cased to a depth of approximately 100 m, and the annulus outside the PVC casing is filled with cement, except for three depth intervals where instruments are installed. For these depths (50, 65, and 90 m), the annulus was filled with gravel (SINTEF, 2025). The monitoring boreholes are located 10 meters from the injection well (M3 and M4) and 16.5 m from the injection well (M1 and M2). Fiber-optic cables, including linear (LIN) and helically wounded cable (HWC) single-mode fibers for DAS measurements, are permanently installed in each of the four monitoring wells, clamped to the outside of the casing. The borehole annulus is filled with cement and gravel. The LIN and HWC fibers are connected to form one loop for DAS interrogation. Previous results published by Koedel et al. (2024) show that the performance of a linear optical fiber is superior to the helical fiber ones if one uses a SV source.

In this experiment, a SV-wave borehole source type BIS-SV was deployed in monitoring well M4 and moved at 1 m steps downward to the bottom of the borehole. The SV-wave source generates a high amount of S-waves and a small amount of P-waves too and can operate in dry or water-filled boreholes. During operation, the source is coupled to the borehole wall by an inflatable packer loaded with an overpressure of about 1.5 bar relative to the water level. Inside the borehole source, two copper coils that can be operated sequentially accelerate a stainless steel weight of about 2.5 kg to produce an “up” and “down” strike. The outside tube is made of a special lightweight carbon fiber to maintain

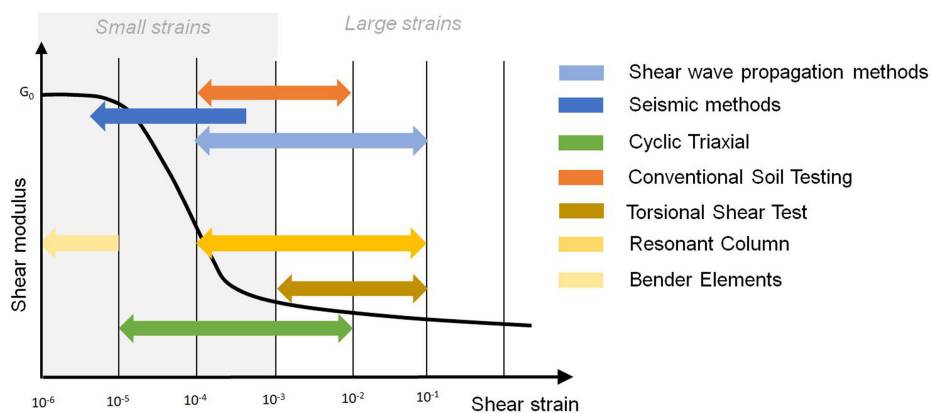


Figure 2. Classification of the test techniques in different strain ranges.

an overall low source weight compared to the accelerated inside weight. At each source depth, an “up” and “down” strike was performed, and signals were stacked five times for each direction. Data acquisition was conducted using an OptoDAS interrogator developed by Alcatel Submarine Networks. Using a frequency-swept interrogation technique, the system was configured with a spacing of 1.02 m between measurement points, a gauge length of 2.04 m, and an effective sampling frequency of 5 kHz. The approximately 2.3-km fiber loop connected to the interrogator included a straight fiber subloop extending along all monitoring boreholes, three helically wound fiber loops covering boreholes M1, M2, and M3, and an additional straight fiber loop placed within the injection borehole.

To compare the seismic data recorded with DAS, a conventional borehole geophone was deployed in borehole M3 and moved parallel to the borehole source to acquire a complete crosswell data set. A standard engineering seismograph was used to record the conventional data simultaneously. The test procedure is similar to the crosswell test procedure described under ASTM 4428 (ASTM, 2025).

During the S-wave survey, we measured an average speed of 3–4 min per shot depth. This includes coupling the SV-wave source at a certain depth, firing five shots up and five shots down, and decoupling the source. This is about twice the time required for a simple P-wave survey, with about 1–2 min per shot depth, where one only has to move the P-source to a new position and no clamping is required. In any case, a conventional survey requires water-filled boreholes for the sparker source and hydrophones. This is not necessary for the S-wave source and not necessary for the DAS where the fiber can be cemented outside the casing.

Data processing and results. The excitation was performed in two directions to take advantage of the polarization characteristics of the S-wave. When selecting the first arrivals, it is clear that the first signal to arrive is usually the P-wave, followed by the S-wave. The arrival times of the conventional S-wave data were determined manually on the vertical geophone channel. This was done by combining the “up” and “down” strikes using different colors and picking the arrival times at the times when the opposite polarization started (see Figure 4). The arrival times are based on the locations of the source and

the geophone receiver, which are positioned at the same depth. To calculate the seismic velocity, the borehole deviations of the two boreholes were taken into account. Nonetheless, the two boreholes demonstrate minimal variations, with deviations of no more than 40 cm at their bottom-hole location.

Processing the DAS records to compute a seismic tomogram consists of several steps. First, the relevant fiber intervals are cropped. Then, the measurement geometry is assigned to each shot location of the S-wave source and receiver locations, i.e., the recording locations along the DAS fiber. Third, the arrival times of the S-wave for all shot and receiver positions were determined manually. The data were filtered with a second-order Butterworth bandpass filter set to a frequency range of 150–2000 Hz. To identify the S-wave arrivals, we took into account the different polarization characteristics visible in the DAS recordings for the “up” and “down” directions of impact (see Figure 5). In total, approximately 6800 S-wave traveltimes were determined. GeotomCG software (Geotom, LLC) was used for tomographic inversion. Processing time, i.e., the manual traveltimes picking of a DAS record with about 95 traces, took about 2–7 min. For comparison, it took about 2–3 min for a conventional P-wave tomography record with 48 channels. Of course, the processing times may vary depending on signal quality.

To convert the traveltimes into an areal distribution of seismic velocities, the subsurface has to be divided into 3D cells as a model. The cell sizes for inversion were chosen to be 1 m × 1 m × 0.9 m in the x -, y -, and z -directions. During the inversion procedure, seismic rays with source–receiver angles exceeding 35° were neglected. This procedure eliminates rays with steep angles, which often lead to artifacts as they tend to bend outside the observation plane.

The inversion started using a horizontally layered model generated by using only seismic rays within an incident angle of $\pm 5^\circ$ to the horizontal. The borehole profiles obtained from drilling indicate a horizontally layered soil structure. The traveltimes associated with these rays are based on the locations of the seismic source and receiver at the same depth level. Thus, we additionally used the filtered subset of the DAS traveltimes to compare them with the conventional crosswell traveltimes of the borehole geophone data (see Figure 6).

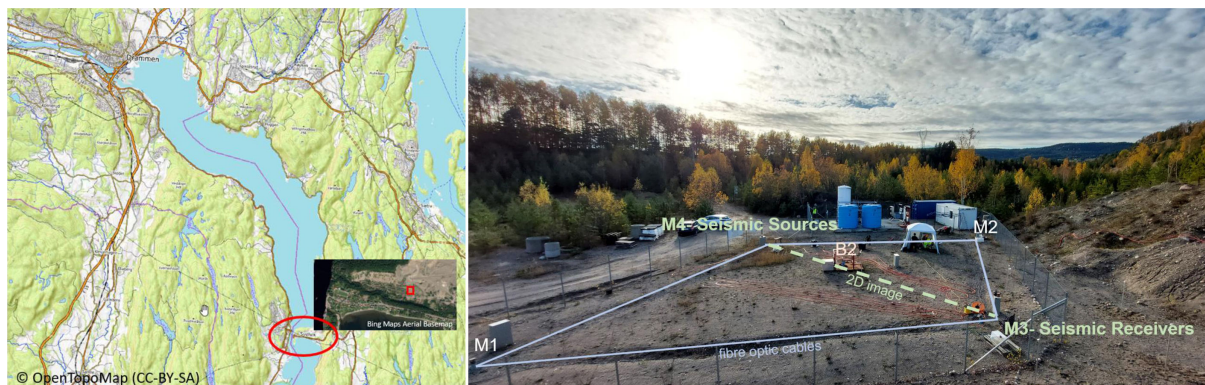


Figure 3. Location of Svelvik CO₂ field lab (left, Source: Open Topo Map, Bing Aerial) and the Svelvik CO₂ field lab with injection (center, B2) and monitoring wells (M1–M4) (right).

The comparison of the two crosswell data sets shows that the DAS and conventional data traveltimes for the horizontal rays are in very good agreement. There is a clear trend of shorter

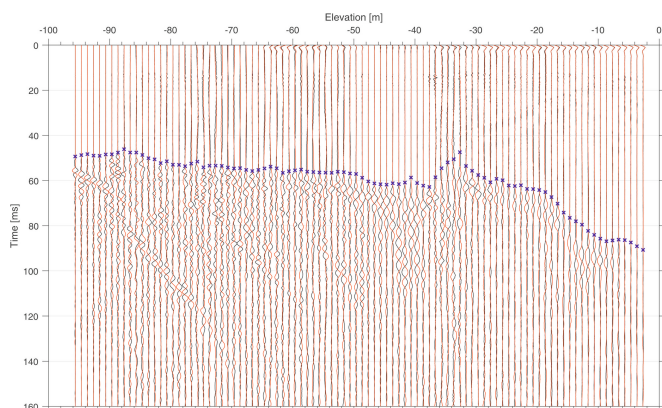


Figure 4. Conventional seismic crosswell survey record showing the vertical borehole geophone channel, with the opposite polarization direction in red and black. Traveltime picks are displayed as blue crosses.

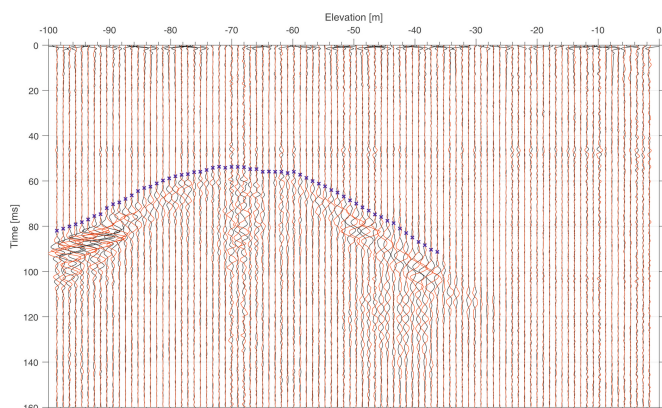


Figure 5. Example of a DAS seismic crosswell record for a shot depth of 69 m with the opposite polarization direction. Traveltime picks are displayed as blue crosses.

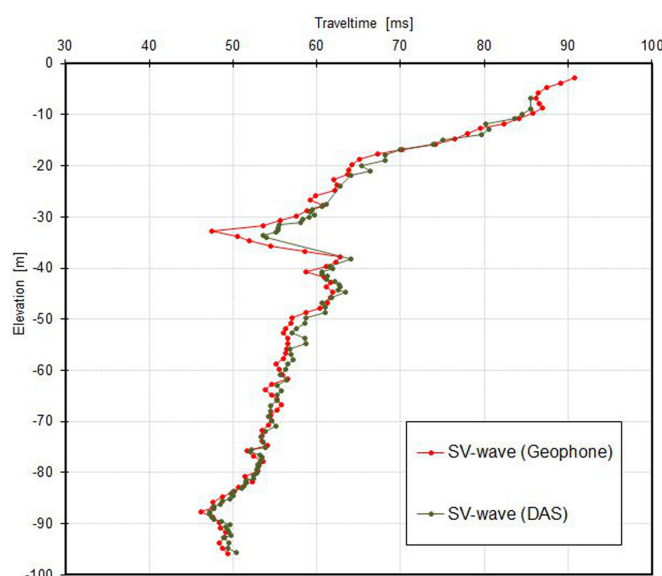


Figure 6. Comparison of conventional S-wave traveltimes and the filtered DAS data subset for horizontal rays.

traveltimes or higher seismic S-wave velocities with increasing depth. Generally, it is observed that the traveltimes of the DAS data are slightly late compared to those of the conventional data. This discrepancy may be explained by the averaging effect caused by the gauge length of the DAS acquisition.

The inversion, based on the horizontally layered starting model, was finally carried out using a curved ray inversion with 10 simultaneous iterative reconstruction technique iterations implemented in the GeotomCG software. The traveltime residuals obtained after the final iteration suggest the presence of a minor degree of anisotropy in the SV wave data. Although GeotomCG offers the option to account for SV anisotropy during inversion, this feature was not explored further in the present study. Tests revealed that, while taking the small anisotropy into account during inversion, it had minimal impact on the final calculated velocity model. It is acknowledged that, in combination with potential picking inaccuracies and the averaging effects introduced by the gauge length, anisotropy may contribute to uncertainties in the inversion process and influence the final results. Particular care was taken during traveltime picking to preserve a consistent wavelet phase throughout the data set. Misidentifying the signal phase can result in systematic traveltime errors that may manifest as artificial anisotropy dependent on ray path orientations.

Figure 7a shows the tomographic inversion result of the S-wave velocity distribution obtained from the linear DAS data after 10 iterations. The seismic velocity ranges from approximately 150 m/s in the upper part to approximately 500 m/s below 60 m, and the S-wave tomogram indicates a high-velocity zone at a depth of 30–35 m. This high-velocity zone was already confirmed by borehole logging data and a P-wave seismic tomography test performed during a CO₂ injection experiment in 2021 (Koedel et al., 2022, 2024). Figure 7b shows the P-wave tomogram acquired in 2021 at depths between 30 and 77 m using two conventional 24-channel hydrophone strings spaced 1 m. Comparing the S-wave velocity structures to the P-wave structures reveals significant differences. The P-wave tomogram shows a more or less horizontal layering of low- and high-velocity zones, whereas the S-wave tomogram displays a homogeneous area of moderately high S-wave velocities and an area of higher velocities beginning at 60 m toward the bottom. Thus, apart from the high-velocity zone at about 35 m depth, no similarities can be observed. The relatively homogeneous S-wave velocity structure observed at the site contrasts with the more distinctly layered P-wave velocity profile. This discrepancy may be attributed to the differing sensitivities of the two wave types. S-wave propagation is primarily influenced by the soil's shear stiffness, whereas P-wave propagation is mainly governed by the compressional bulk modulus. This observation is consistent with a low R^2 value of 0.196 calculated through a linear correlation between P- and S-wave traveltimes obtained from the conventional crosswell test (see Figure 8).

Thus, the typical assumption of a good linear correlation between P- and S-wave velocities, which is sometimes used to convert a P-wave tomogram into an S-wave distribution, is not

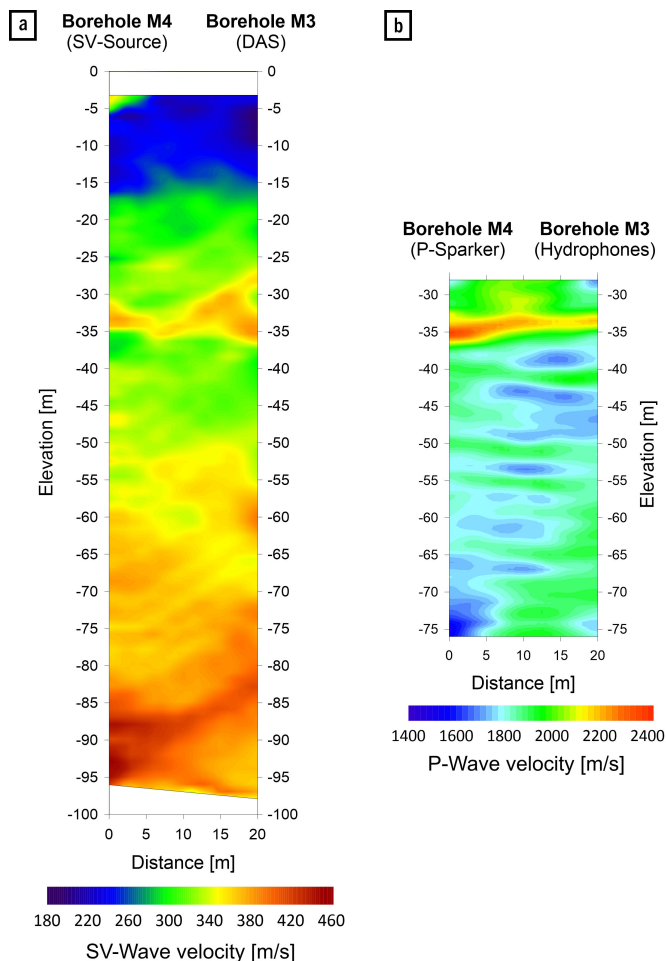


Figure 7. (a) S-wave seismic tomogram using DAS and the SV-wave source (BIS-SV, Geotomographie GmbH) for a depth interval between 0 and 100 m. (b) P-wave seismic tomogram with a P-wave source (SBS42, Geotomographie GmbH) and hydrophone strings (BHC5, Geotomographie GmbH) obtained during the CO₂ injection experiment in 2021 for a depth between 30 and 77 m.

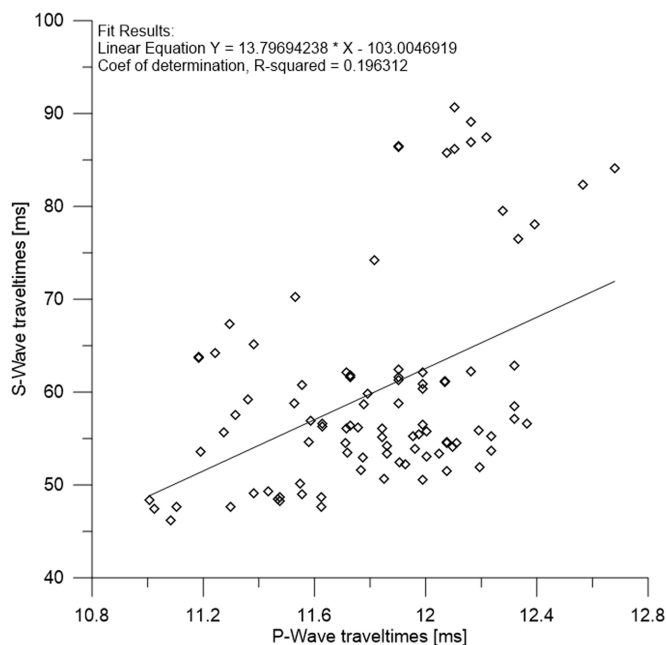


Figure 8. Correlation between P- and S-wave traveltimes of the conventional crosswell data.

justified for the presented data, at least. Furthermore, it can be seen that the variability of the P-wave traveltimes is approximately 13%, whereas it is about 50% for the S-wave traveltimes. This shows that S-waves are more sensitive to soil variations than P-waves.

To obtain a distribution of soil stiffness, i.e., the small-strain shear modulus (G_0) is calculated using the S-wave velocity (V_S) and the soil density (ρ) by using the following equation: $G_0 = \rho V_S^2$ (e.g., Guadalupe et al., 2013). Essentially, the square of the S-wave velocity, multiplied by the density, represents the material's resistance to shear deformation at small strain levels. Assuming a density value of 2000 kg/m³, a dynamic shear stiffness map can be calculated and is shown in Figure 9.

The shear stiffness values obtained through seismic measurements are related to the highest stiffness value at a very low strain rate (Benz, 2007; Asslan, 2008). Because the relationship between shear stiffness and strain rate is highly nonlinear, this value is the basis for all numerical calculations of soil deformation behavior under cyclic loading. Other tests, such as the flat dilatometer test or standard penetration test, operate at higher strain rates than seismic measurements and can yield lower stiffness values for soils (Marchetti et al., 2016; Marchetti, 2020). Increased strain rates, such as those experienced during an earthquake, can cause soil to lose stiffness, resulting in liquefaction. Especially, the rapid and repeated shaking during an earthquake (high strain rates) causes a build-up of pressure within the water-filled pore spaces of the soil, leading to a loss of strength and the potential for liquefaction (Ben-Zeev et al., 2023).

Discussion

S-wave tomography has traditionally been limited by technical constraints, primarily due to the need for numerous mechanically coupled geophones to provide sufficient angular coverage. The DAS-based approach presented here overcomes these limitations with a full receiver coverage from top to the bottom of the borehole. The field experiment demonstrated that acquisition with DAS can be completed within 3–4 min per shot depth, only slightly longer than P-wave surveys, but without the need for water-filled boreholes or complex equipment repositioning. In addition, for this experimental layout, DAS records contain approximately 95 traces per shot, nearly doubling the spatial resolution of conventional 2×24 -channel hydrophone systems for P-wave tomography. This higher data density improves inversion quality and reduces the number of shot positions required to achieve a sufficient coverage if only 24 or 48 channels are used and strings need to be shifted, resulting in significant time and cost savings. It is estimated that survey costs can be reduced by 30%–50% using this method, particularly in applications involving repeated measurements or long-term monitoring.

Importantly, the tomographic results revealed that the structures imaged by S-wave and P-wave data are significantly different, with a low correlation ($R^2 = 0.196$) observed between their respective traveltimes. This indicates that S-wave velocity

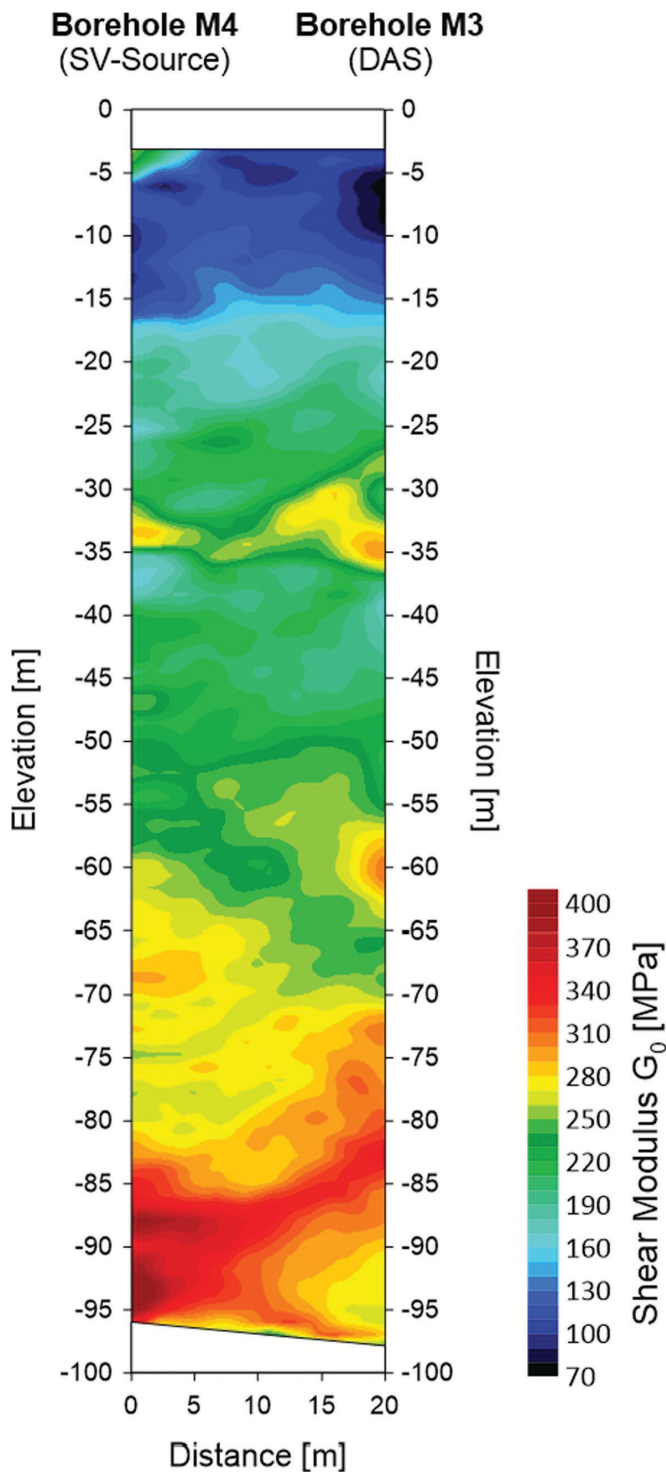


Figure 9. Distribution of shear modulus G_0 at small strain obtained from the S-wave tomogram velocities and an assumed average material density.

distributions cannot be reliably derived from P-wave data and highlights the complementary nature of both wave types.

Conclusion

This study demonstrates the feasibility and advantages of using DAS in combination with a borehole S-wave

source for high-resolution S-wave crosswell tomography. Linear optical fibers, which exhibit optimal sensitivity to vertically polarized shear waves, were used to acquire dense and high-quality data sets with improved spatial coverage compared to conventional geophones. The SV-wave source (BIS-SV) is specifically designed to generate vertically polarized shear waves, which are well aligned with the directional sensitivity of linear DAS fibers installed along vertical boreholes. When used in conjunction with DAS, the SV source enabled the efficient acquisition of high-resolution S-wave data without the need for conventional borehole geophones. Due to the predominantly vertical particle motion of SV-waves, optimal signal coupling was achieved along the vertical axis of the fiber, enhancing the clarity and consistency of the recorded waveforms. This source-fiber alignment significantly increased the number of usable traveltime picks, thereby improving the spatial resolution and robustness of the tomographic inversion.

The major advantage of S-wave tomography is that one of the most important geotechnical parameters, soil stiffness, can be directly derived assuming a known soil density. This provides a crucial parameter for geotechnical applications, especially under small strain conditions. Because shear stiffness plays a central role in the evaluation of soil behavior under cyclic or dynamic loading, such information is highly valuable for earthquake hazard assessment and geotechnical foundation design. Operational benefits were demonstrated through the integration of DAS and the borehole-coupled SV-wave source. The combination of SV-wave borehole source and DAS enhances resolution and sensitivity in detecting variations in subsurface properties, outperforming traditional geophones in coverage and data resolution. This approach also eliminates the need for water-filled boreholes and complex mechanical coupling, simplifying field procedures and reducing logistical demands. This setup also offers notable cost-efficiency and enables scalable, high-resolution subsurface imaging. As a result, the combined use of DAS and SV-wave sources represents a significant advancement in geotechnical and environmental site characterization, offering substantial benefits for geotechnical engineering, environmental monitoring, and infrastructure development, allowing for scalable, noninvasive investigations with enhanced efficiency and data quality. **TLE**

Acknowledgments

The presented work was funded by the Geo-INQUIRE program (TA1-44-1), and this support is gratefully acknowledged. Geo-INQUIRE program, funded by the European Union (EU, project number 101058518), aims to enhance and integrate geological infrastructures across Europe. It provides researchers with access to various research facilities and data to advance their scientific knowledge of earth processes. The access to the Svelvik CO₂ Field laboratory was provided by SINTEF. The DAS-related part of this work was supported by SINTEF through a strategic internal project (SIP) titled “Crystal Clear.”

Data and materials availability

Data associated with this research are available and can be obtained by contacting the corresponding author.

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References

- Asslan, M., 2008, Factors influencing small-strain stiffness of soils and its determination: Thesis PhD, Bauhaus University Weimar, Professorship Soil Mechanics.
- ASTM, 2018, Standard test methods for downhole seismic testing. ASTM D7400-14: ASTM International, <https://doi.org/10.1520/D7400-14>.
- ASTM, 2025, Standard test methods for crosshole seismic testing of soil and rock, ASTM D4428/D4428M-25, ASTM International. https://doi.org/10.1520/D4428_D4428M-25.
- Atkinson, J. H., 2000, Non-linear soil stiffness in routine design: *Geotechnique*, **50**, no. 5, 487–508, <https://doi.org/10.1680/geot.2000.50.5.487>.
- Benz, T., 2007, Small-strain stiffness of soils and its numerical consequences: Thesis PhD, Faculty of Civil and Environmental Engineering, University of Stuttgart.
- Ben-Zeev, S., L. Goren, R. Toussaint, and E. Aharonov, 2023, Drainage explains soil liquefaction beyond the earthquake near-field: *Nature Communications*, **14**, no. 1, 5791, <https://doi.org/10.1038/s41467-023-41405-4>.
- Boullenger, B., A. Butcher, B. Paap, V. Vandeweyer, and A. L. Stork, 2025, Distributed acoustic sensing, transfer functions, and moment magnitude determination: Application to high-frequency downhole data: *Geophysics*, **90**, no. 3, L75–L84, <https://doi.org/10.1190/geo2024-0059.1>.
- Burland, J. B., 1989, Ninth Laurits Bjerrum memorial lecture: “Small is beautiful”—the stiffness of soils at small strains: *Canadian Geotechnical Journal*, **26**, no. 4, 499–516, <https://doi.org/10.1139/t89-064>.
- Daley, T. M., D. E. Miller, K. Dodds, P. Cook, and B. M. Freifeld, 2016, Field testing of modular borehole monitoring with simultaneous distributed acoustic sensing and geophone vertical seismic profiles at Citronelle, Alabama: *Geophysical Prospecting*, **64**, no. 5, 1318–1334, <https://doi.org/10.1111/1365-2478.12324>.
- Eliasson, P., C. Ringstad, A. Grimstad, M. Jordan, and A. Romdhane, 2018, Svelvik CO2 field lab: Upgrade and experimental campaign: Fifth CO2 Geological Storage Workshop, **2018**, 1–5, <https://doi.org/10.3997/2214-4609.201802973>.
- Fichtner, A., B. L. N. Kennett, V. C. Tsai, C. H. Thurber, A. J. Rodgers, C. Tape, N. Rawlinson, et al., 2024, Seismic tomography 2024: *Bulletin of the Seismological Society of America*, **114**, no. 3, 1185–1213, <https://doi.org/10.1785/0120230229>.
- Guadalupe, Y., C. D. P. Baxter, and M. S. R. Sharma, 2013, Measuring shear wave velocity in laboratory to link small- and large-strain behavior of soils: *Transportation Research Record*, **2335**, no. 1, 79–88, <https://doi.org/10.3141/2335-09>.
- Halladay, A., V. Orpeza Bacci, S. O'Brien, and K. Hindriks, 2018, November, Results from the second monitor DAS VSP at quest CCS, in EAGE Fifth CO2 Geological Storage Workshop, 21–23, <https://doi.org/10.3997/2214-4609.201802981>.
- Hussien, M. N., and M. Karray, 2015, Shear wave velocity as a geotechnical parameter: An overview: *Canadian Geotechnical Journal*, **53**, no. 2, 252–272, <https://doi.org/10.1139/cgj-2014-0524>.
- Jamiolkowski, M., D. C. F. LoPresti, and O. Pallara, 1995, Role of in-situ testing in geotechnical earthquake engineering: *Proceedings of 3rd International Conference on Recent Advances in Geotechnical Earthquake Engineering and Soil Dynamics*, St. Louis 1995, vol III, 1523–1546.
- Koedel, U., T. Fechner, and A. Stork, 2022, Project DIGIMON Deliverable 2.9: Project report on seismic tomography data interpretation and conceptual model for integrating DAS into borehole seismic tomography surveying, <https://core.ac.uk/download/568283838.pdf>, accessed 10 January 2025.
- Koedel, U., A. L. Stork, A. David, and T. Fechner, 2024, Evaluating distributed acoustic sensing for crosswell seismic surveys with helical and linear fibers using conventional P-, SH-, and SV-wave sources: *The Leading Edge*, **43**, no. 11, 726–734, <https://doi.org/10.1190/tle43110726.1>.
- Marchetti, D., 2020, Applications of Dilatometer (DMT) and Seismic Dilatometer (SDMT) in Geotechnical Engineering, Presentation at Stuttgarter Geotechnik-Seminar on November 20, 2020, available at, https://www.igs.uni-stuttgart.de/geotechnik-seminar/Programm_aktuell/201116_Marchetti.pdf, accessed 17 May 2025.
- Marchetti, S., D. Marchetti, and P. Monaco, 2016, Flat dilatometer (DMT): Applications and recent developments: 1st IMEKO TC-4 International Workshop on Geotechnics, , accessed 17–18 March 2026, BEnevento, Italy, <https://www.imeko.org/publications/tc4-Geo-2016/IMEKO-TC4-GEO-2016-23.pdf>, accessed 18 May 2025.
- Meyer, V., T. Larkin, and M. Pender, 2005, The shear strength and dynamic shear stiffness of some New Zealand volcanic ASH soils: *Soils and Foundations*, **45**, no. 3, 9–20, https://doi.org/10.3208/sandf.45.3_9.
- Minto, J., B. Marion, M. Shafiq, and A. Nalonnil, 2012, Next generation borehole seismic source: Dual-wavefield downhole vibrator system: ASEG Extended Abstracts, 22nd Geophysical Conference, 1, <https://doi.org/10.1071/ASEG2012ab143>.
- Nakajima, T., T. Miyoshi, T. Hashimoto, and Z. Xue, 2022, Case study on amplitude and phase response comparison between DAS and 3C geophone VSP surveys at a test site in Japan: *Frontiers in Earth Science*, **10**, 991964, <https://doi.org/10.3389/feart.2022.991964>.
- Nalonnil, A., B. Marion, and J. Minto, 2013, March 26, Next generation borehole seismic: Dual-wavefield vibrator system: International Petroleum Technology Conference, <https://doi.org/10.2523/IPTC-16870-MS>.
- Paasche, H., M. Ohrnberger, U. Werban, T. Fechner, J. Tronicke, F. Krüger, and P. Dietrich, 2011, Multi-scale S-wave tomography for exploration and risk assessment of development sites (MuSaWa), in U. Münch, ed., *Geotechnologien Science Report*, 24–34, <https://doi.org/10.2312/GFZ.gt.18.03>.
- Pereira, A. M., and M. Jones, 2010, *Fundamentals of borehole seismic technology*. Texas, USA: Schlumberger.
- Podgornova, O., S. Ali, T. Lei, P. Li, M. Johnson, L. Lee, Z. Jiao, et al, 2023, Anisotropic elastic full-waveform inversion for crosswell seismic data for Wyoming CarbonSAFE project: Third International Meeting for Applied Geoscience & Energy, SEG/AAPG, 225–229, <https://doi.org/10.1190/image2023-3914313.1>.
- Rammah, K. I., M. A. Ismail, J. Costa, and M. V. Riccio Filho, 2022, A new seismic tomography system for geotechnical centrifuges: *Soils and Rocks*, **46**, no. 1, e2023000922, <https://doi.org/10.28927/SR.2023.000922>.
- Rawlinson, N., A. Fichtner, M. Sambridge, and M. K. Young, 2014, Seismic tomography and the assessment of uncertainty: *Advances in Geophysics*, **55**, 1–76, <https://doi.org/10.1016/bs.agph.2014.08.001>.
- SINTEF, 2025. ECCSEL svelvik CO2 field lab, Report, <https://www.sintef.no/contentassets/4c8ac1eb1c844e30a7b05558863f8846/svelvik-co2-field-lab---brochure-2024.pdf>, accessed 18 May 2025.

- Stewart, R. R., 1991. Exploration seismic tomography: Fundamentals, Society of Exploration Geophysicists, <https://doi.org/10.1190/1.9781560802372>.
- Tatsuoka, F., R. Jardine, D. Lo Presti, H. Di Benedetto, and T. Kodaka, 1999, Characterizing the pre-failure deformation properties of geomaterials: Proc. 14th International Conference on Soil Mechanics and Foundation Engineering (Hamburg), <https://www.issmge.org/publications/publication/theme-lecture-characterizing-the-pre-failure-deformation-properties-of-geomaterials>, accessed 16 May 2025.
- von Ketelhodt, J. K., T. Fechner, M. S. D. Manzi, and R. J. Durrheim, 2017, Elastic parameters from compressional and shear wave tomographic survey: A case study from Kuala Lumpur, Malaysia: Journal of Environmental and Engineering Geophysics, **22**, no. 4, 427–434, <https://doi.org/10.2113/JEEG22.4.427>.
- Willis, M. E., 2022, Distributed acoustic sensing for seismic measurements – what geophysicists and engineers need to know: Society of Exploration Geophysicists, Distinguished Instructor Series, <https://doi.org/10.1190/1.9781560803850>.
- Willis, M. E., J. Ajo-Franklin, and B. Roy, 2017, Introduction to this special section: Geophysical applications of fiber-optic distributed sensing: The Leading Edge, **36**, no. 12, 973–974, <https://doi.org/10.1190/tle36120973.1>.
- Wu, X., M. E. Willis, W. Palacios, A. Ellmuthaler, O. Barrios, S. Shaw, and D. Quinn, 2017, Compressional- and shear-wave studies of distributed acoustic sensing acquired vertical seismic profile data: The Leading Edge, **36**, no. 12, 987–993, <https://doi.org/10.1190/tle36120987.1>.
- Wuestefeld, A., and W. Weinzierl, 2020, Design considerations for using distributed acoustic sensing for crosswell seismics: A case study for CO₂ storage: Geophysical Prospecting, **68**, no. 6, 1893–1905, <https://doi.org/10.1111/1365-2478.12965>.
- Yurikov, A., K. Tertyshnikov, S. Yavuz, P. Shashkin, R. Isaenkov, E. Sidenko, S. Glubokovskikh, et al., 2022, Seismic monitoring of CO₂ geosequestration using multi-well 4D DAS VSP: Stage 3 of the CO₂CRC Otway project: International Journal of Greenhouse Gas Control, **119**, 103726, <https://doi.org/10.1016/j.ijggc.2022.103726>.